

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I U Fayaz Ahmad (192212182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1. DOCTOR SHIFT ASSIGNMENT USING BACKTRACKING SEARCH

Aim

To assign three doctors (D1, D2, D3) to three different shifts (Morning, Afternoon, Night) using the **Backtracking Search Algorithm** while satisfying all the given constraints.

Problem Statement

A hospital needs to assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)**.

Constraints

1. D1 cannot work **Night** shift.
2. D2 must work before D3 (**Morning < Afternoon < Night**).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work **Morning** shift.
5. Every doctor must be assigned exactly one shift.

Variables

- D1
- D2
- D3

Domain

Each doctor can be assigned one of the following shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Initial State

No doctor has been assigned.

Doctor Shift

D1 —
D2 —
D3 —

Backtracking Search Process

Step 1: Assign D1

Try **Morning**

D1 → Morning ✓

Constraint Check:

- D1 is not assigned Night ✓

Current Assignment

Doctor Shift

D1 Morning
D2 —
D3 —

Step 2: Assign D2

Morning is already occupied.

Try Afternoon.

D2 → Afternoon ✓

Constraint Check:

- One doctor per shift ✓
- D2 is before D3 (possible) ✓

Current Assignment

Doctor Shift

D1 Morning
D2 Afternoon
D3 —

Step 3: Assign D3

Morning ✗

Constraint violated:

- D3 cannot work Morning.

Try Afternoon ✗

Constraint violated:

- Afternoon already assigned to D2.

Try Night ✓

Constraint Check:

- D3 not in Morning ✓
- D2 works before D3 (Afternoon < Night) ✓
- Only one doctor per shift ✓

Current Assignment

Doctor Shift

D1 Morning
D2 Afternoon
D3 Night

Constraint Verification

Constraint 1

D1 cannot work Night

D1 → Morning ✓

Satisfied.

Constraint 2

D2 must work before D3

D2 → Afternoon

D3 → Night

Afternoon comes before Night ✓

Satisfied.

Constraint 3**Only one doctor per shift**

Morning → D1

Afternoon → D2

Night → D3

No duplication ✓

Satisfied.

Constraint 4**D3 cannot work Morning**

D3 → Night ✓

Satisfied.

Constraint 5**Each doctor assigned exactly one shift**

D1 ✓

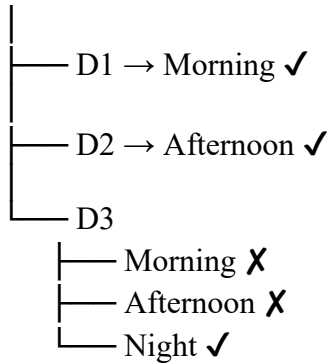
D2 ✓

D3 ✓

Satisfied.

Backtracking Search Tree

Start



Final Valid Schedule

Shift	Doctor
-------	--------

Morning	D1
---------	----

Afternoon	D2
-----------	----

Night	D3
-------	----

Result

Thus, using the **Backtracking Search Algorithm**, a valid assignment satisfying all the given constraints is obtained.

Final Schedule:

- **Morning → D1**
- **Afternoon → D2**
- **Night → D3**

All constraints are successfully satisfied, making this the correct solution.

2. ROBOT GRID NAVIGATION USING BREADTH-FIRST SEARCH (BFS)

Aim

To determine the optimal path for a robot to move from the **Start (S)** position to the **Goal (G)** position in a **5 × 5 grid** using the **Breadth-First Search (BFS)** algorithm while avoiding obstacles.

Problem Statement

A robot operates in a **5 × 5 grid environment**.

The robot must move from **Start (S)** to **Goal (G)** under the following constraints:

Constraints

1. Movement allowed only in **Up, Down, Left, Right**
 2. Cost of each move = **1**
 3. Robot cannot move through obstacles
 4. Use **Manhattan Distance** heuristic
-

Grid Representation

C1 C2 C3 C4 C5

R1 S • X • •

R2 • X • X •

R3 • • • X •

R4 X X • • •

R5 • • • X G

Where:

- S = Start (R1,C1)
 - G = Goal (R5,C5)
 - X = Obstacle
 - • = Free Cell
-

Initial State

Start Position = **(1,1)**

Goal Position = **(5,5)**

Queue = [(1,1)]

Visited = {(1,1)}

Manhattan Distance Formula

[
 $h(n) = |x_{\{goal\}} - x| + |y_{\{goal\}} - y|$
]

Goal = (5,5)

Example:

For Start (1,1)

[
 $h = |5-1| + |5-1| = 8$
]

Breadth-First Search (BFS) Steps

Step 1

Expand:

(1,1)

Possible moves

- Right $\rightarrow$ (1,2) ✓
- Down $\rightarrow$ (2,1) ✓

Queue

[(1,2),(2,1)]

Visited

(1,1),(1,2),(2,1)

Step 2

Expand

(1,2)

Possible moves

- Right $\rightarrow$ Obstacle ✗
- Down $\rightarrow$ Obstacle ✗

No new node added.

Queue

[(2,1)]

Step 3

Expand

(2,1)

Possible moves

- Down $\rightarrow$ (3,1) ✓

Queue

[(3,1)]

Visited

(3,1)

Step 4

Expand

(3,1)

Possible moves

- Right $\rightarrow$ (3,2) ✓

Queue

[(3,2)]

Step 5

Expand

(3,2)

Possible moves

- Right $\rightarrow$ (3,3) ✓

Queue

[(3,3)]

Step 6

Expand

(3,3)

Possible moves

- Down $\rightarrow$ (4,3) ✓
- Up $\rightarrow$ (2,3) ✓

Queue

[(4,3),(2,3)]

Step 7

Expand

(4,3)

Possible moves

- Right $\rightarrow$ (4,4) ✓
- Down $\rightarrow$ (5,3) ✓

Queue

[(2,3),(4,4),(5,3)]

Step 8

Expand

(4,4)

Possible moves

- Right $\rightarrow$ (4,5) ✓

Queue

[(5,3),(4,5)]

Step 9

Expand

(4,5)

Possible moves

- Down $\rightarrow$ Goal (5,5) ✓

Goal Found.

BFS Node Expansion Table

Step Expanded Node Queue After Expansion

1	(1,1)	(1,2), (2,1)
2	(1,2)	(2,1)
3	(2,1)	(3,1)
4	(3,1)	(3,2)
5	(3,2)	(3,3)
6	(3,3)	(4,3), (2,3)
7	(4,3)	(2,3), (4,4), (5,3)
8	(4,4)	(5,3), (4,5)
9	(4,5)	(5,5)

Optimal Path

S

↓

(2,1)

↓

(3,1)



(3,2)



(3,3)



(4,3)



(4,4)



(4,5)



G

Or,

(1,1)

→ (2,1)

→ (3,1)

→ (3,2)

→ (3,3)

→ (4,3)

→ (4,4)

→ (4,5)

→ (5,5)

Path Cost

Each movement costs 1.

Total Moves = 8

Total Cost = $8 \times 1 = 8$

Result

Thus, using the **Breadth-First Search (BFS)** algorithm, the robot successfully reaches the **Goal (G)** from the **Start (S)** while avoiding obstacles. The **optimal path** consists of **8 moves**, and the **total path cost is 8**. The Manhattan Distance heuristic is used to estimate the distance to the goal, while BFS guarantees the shortest path in terms of the number of moves.

3. AUTONOMOUS RESCUE ROBOT – CONSTRAINT-BASED PROBLEM SOLVING

Aim

To analyze the operation of an autonomous rescue robot in a disaster-affected building and develop a suitable AI-based solution using an appropriate search strategy while satisfying all the given constraints.

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot starts at **Start (S)** and must reach the survivor located at **Goal (G)**. The robot operates in a **partially observable environment** where some paths are blocked, and it has limited sensing capability. The robot must find a safe path with the minimum possible cost.

Given Constraints

1. The robot can move only **Up, Down, Left, and Right**.
 2. Some cells are blocked (obstacles) and cannot be crossed.
 3. The robot can observe only adjacent cells.
 4. Every movement has a cost of **1**.
 5. Entering a risky zone has an additional cost of **2**.
 6. The robot must minimize the total cost.
 7. The robot must continuously update its knowledge while moving (Online Search).
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(a) Constraint Analysis

1. Movement Constraint

The robot is allowed to move only in four directions:

- Up
- Down
- Left
- Right

This limits the possible paths and simplifies navigation.

2. Obstacle Constraint

Some rooms are blocked due to debris or collapsed structures.

The robot must avoid these blocked cells and search for an alternative path.

3. Limited Sensing Capability

The robot can detect only the adjacent rooms.

Since it cannot view the entire environment at once, it must make decisions based on local information.

4. Movement Cost

Each movement from one cell to another costs **1**.

The robot should minimize the number of movements.

5. Risk Zone Cost

If the robot enters a risky area, an additional cost of **2** is added.

For example:

- Normal Cell = 1
- Risky Cell = 1 + 2 = 3

Therefore, the robot should avoid risky zones whenever possible.

6. Online Knowledge Update

As the robot explores the building, it updates its internal map using newly discovered information.

This enables the robot to choose safer and more efficient paths.

(b) Solution Using Uniform Cost Search (UCS) / Online Search Agent

Chosen Search Strategy

Uniform Cost Search (UCS) with an **Online Search Agent** is the most suitable approach.

Reason

- UCS always expands the node with the lowest cumulative cost.
 - It guarantees the least-cost path.
 - The Online Search Agent updates the robot's knowledge dynamically when new obstacles or risky zones are detected.
-

Solution Steps

Step 1

The robot starts at **S**.

Current Cost = **0**

Step 2

Observe adjacent cells.

- Ignore blocked cells.
 - Identify safe and risky cells.
-

Step 3

Move to the neighboring cell with the **lowest cumulative cost**.

Update Total Cost.

Step 4

Repeat the process.

Whenever a new obstacle or risky zone is discovered:

- Update the internal map.
 - Recalculate the safest path.
-

Step 5

Continue searching until the survivor (**G**) is reached.

Example Cost Calculation

Path	Cost
------	------

$S \rightarrow A \rightarrow B \rightarrow G$	6
---	---

$S \rightarrow C \rightarrow D \rightarrow G$	5
---	---

$S \rightarrow \text{Risk Zone} \rightarrow G$	8
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Uniform Cost Search selects:

$S \rightarrow C \rightarrow D \rightarrow G$

because it has the **minimum total cost (5)**.

(c) Application of AI Concepts

1. Intelligent Agent

The rescue robot is an **Intelligent Agent** because it:

- Perceives the environment using sensors.
- Processes information.

- Makes decisions.
 - Performs actions to achieve its goal.
 - Continuously updates its knowledge.
-

2. Rationality

A rational agent always selects the action that maximizes its performance.

The rescue robot behaves rationally by:

- Choosing the safest path.
 - Minimizing travel cost.
 - Avoiding risky zones.
 - Reaching the survivor efficiently.
-

3. Environment Type

The rescue robot operates in the following environment:

Property	Description
Partially Observable	Robot can observe only nearby cells.
Dynamic	Environment may change due to collapsing structures.
Sequential	Current decisions affect future actions.
Stochastic	Unexpected events may occur.
Discrete	Robot moves between grid cells.

(d) Justification of the Chosen Approach

Uniform Cost Search is the most suitable algorithm because:

- It finds the **minimum-cost path**.
- It considers different movement costs.
- It avoids expensive risky zones whenever possible.
- It guarantees an optimal solution.
- It works effectively with an Online Search Agent that updates knowledge dynamically as new information becomes available.

Therefore, UCS combined with an Online Search Agent enables the rescue robot to navigate safely and efficiently in a disaster-affected environment.

Conclusion

The autonomous rescue robot successfully applies AI concepts to navigate a partially observable environment. By using **Uniform Cost Search (UCS)** together with an **Online Search Agent**, the robot continuously updates its knowledge, avoids obstacles and risky zones, minimizes the total navigation cost, and reaches the survivor safely.

Result

Thus, the rescue robot efficiently locates the survivor by satisfying all the given constraints.

The **Uniform Cost Search (UCS)** algorithm, supported by an **Online Search Agent**, provides the optimal and safest path while minimizing the total travel cost, making it the most suitable approach for this constraint-based problem.